EXPERIMENT 10:

Illustrate the concept of inter-process communication using message . queue with

a C program

Online C Compiler

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Run

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/ipc.h>
#include <sys/msg.h>

struct message {
    long msg_type;
    char msg_text[100];
};

int main() {
    key_t key = ftok("msgqfile", 65);

    int msgid = msgget(key, IPC_CREAT | 0666);
    if (msgid == -1) {
        perror("msgget");
        exit(EXIT_FAILURE);
    }

    struct message msg;
    msg.msg_type = 1;

    strcpy(msg.msg_text, "Hello, message queue!");

    if (msgsnd(msgid, (void*)&msg, sizeof(msg.msg_text), IPC_NOWAIT) == -1) {
        perror("msgsnd");
        exit(EXIT_FAILURE);
    }

    printf("Producer: Data sent to message queue: %s\n", msg.msg_text);

    if (msgrcv(msgid, (void*)&msg, sizeof(msg.msg_text), 1, 0) == -1) {
        perror("msgrcv");
        exit(EXIT_FAILURE);
    }

    printf("Consumer: Data received from message queue: %s\n", msg.msg_text);

    if (msgctl(msgid, IPC_RMID, NULL) == -1) {
        perror("msgctl");
        exit(EXIT_FAILURE);
    }

    return 0;
}
```

Output

```
Status: Accepted
Memory Used: 732 byte   Execution Time: 0.003 sec

Producer: Data sent to message queue: Hello, message queue!
Consumer: Data received from message queue: Hello, message queue!
```

Input

Enter your input here...